

System Design Specification: Automated Picking & Sorting System

Version: 5.1 **Status:** Engineering Draft **Architecture:** Dual-System (Main X-Z Manipulator + Standalone Sorting Station)

1. Terms and Definitions

To ensure clarity across engineering and operation teams, the following terminology is enforced throughout this specification:

- **Cabinet:** The static shelving unit or storage rack containing the inventory.
- **Drawer:** (Formerly "Penal") The sliding storage unit extracted from the Cabinet.
 - **Standard Length:** 600 mm.
 - **Variants:**
 - **Type S:** 80 mm width, 15 compartments.
 - **Type M:** 100 mm width, 10 compartments.
 - **Type L:** 190 mm width, 4 compartments.
 - **Type B (Bottle):** 90 mm width, 6 compartments (Special design with flexible leafs/tabs to hold bottles, similar to a car cup holder).
- **Item:** Individual SKUs (Boxes or Bottles) stored inside Drawers.
- **Logistics Container:** The standard transport bin handled by the Main Manipulator for order fulfillment.
- **Uploading Container:** A specialized vertical-slot container used by the Sorting Station to buffer incoming items before distribution.
- **Universal Gripper:** The single, adaptive end-effector driven by a lead screw, capable of gripping both thin boxes and wide bottles without tool changing.

2. System Overview

The solution consists of two coordinated robotic subsystems designed for "Goods-to-Robot" automation:

1. **Main X-Z Manipulator:** A mobile linear tower with dual vertical carriages, responsible for retrieving items from Cabinets and transporting Logistics Containers.
2. **Standalone Sorting Station:** A fixed-position vertical SCARA robot responsible for inducting items from a conveyor, verifying them (QR scan), and loading them into Uploading Containers for the Main Manipulator to distribute.

3. Subsystem A: Main X-Z Dual-Carriage Manipulator

3.1. Mechanical Architecture (X-Axis & Tower)

- **Dimensions:**
 - **Tower Height:** Approx. 2.0 meters.
 - **Railway Length:** Approx. 4.0 meters.
- **Kinematics:** Single mobile tower (Body Frame) traveling on a horizontal rail (X-Axis).
- **Drive System:** Rack & Pinion (Module 2, Spur).
- **Actuation:** 400W Servo (High Inertia Type) via 5:1 Planetary Gearbox.
- **Dynamics:** 1.0 m/s² acceleration.
- **Vertical Infrastructure:** A single mobile tower carrying two independent vertical carriages (A & B).
 - **Shared Guides:** Both carriages ride on the same vertical rails and engage the same rack.
 - **Shared Counterweight Rail:** Counterweights for both carriages travel on a dedicated railway on the **opposite side** of the Body Frame to balance the tower's center of mass.

3.2. Z-Axis A (Container Carriage)

- **Function:** Mobile workstation and container holder.
- **Holding:** Static Mechanical Brake (Zero thermal load holding).
- **Counterweight:** Balanced to

$$Carriage + EmptyContainer + 50$$

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- **Container Jaws:** Electromechanical (Stepper/Lead Screw) dual-jaw clamp (left_jaw_link , right_jaw_link).
- **Work Surface:** Top surface acts as a resting plane for Drawers during extraction.

3.3. Z-Axis B (Manipulator Carriage)

- **Function:** Vertical deployment of SCARA Arm 1.
- **Actuation:** 200W Servo (High Inertia) via 10:1 Gearbox.
- **Counterweight:** Balanced to 100% of "Dead Load" (Carriage + SCARA Base + Arm).
 - *Result:* Gravity neutralized (~ON holding torque).

3.4. SCARA Arm 1 (Mobile Picking)

- **Mount:** Fixed to Z-Axis B Carriage.
- **Reach:** 800mm ($L_1 = 400$, $L_2 = 400$).
- **Joint 1 (Shoulder):** 400W Servo, 40:1-50:1 Hybrid Reduction.
- **Joint 2 (Elbow):** 200W Servo, 20:1-30:1 Reduction.

- **Joint 3 (Wrist):** NEMA 17 Stepper (Low mass).
- **Tool: Universal Lead-Screw Gripper** (See Section 6.10).

4. Subsystem B: Standalone Sorting Station (SCARA Arm 2)

A fixed-base robotic station designed for high-speed induction of items from an inbound conveyor into the storage system.

4.1. Mechanical Architecture (Fixed Column)

- **Structural Column:** A rigid, hollow steel box-section column (e.g., 200x200mm) bolted to the floor.
- **Carriage Guidance:** * Two precision **Linear Guide Rails** mounted on the **Front Face** of the column.
 - Carriage Plate rides on 4 heavy-duty linear blocks.
- **Counterweight Mechanics:**
 - **Strategy:** "Drop-In Cartridge" System.
 - **Construction:** An internal skeleton frame made of **Specialized X-Shape Aluminum Profiles** holding the counterweight guides.
 - **Assembly:** The entire cartridge is pre-assembled and lowered into the hollow column.
 - **Fastening:** Bolted from the **OUTSIDE** of the steel tower using **Specialized Profile Fasteners** (No tapping of aluminum required).
 - **Note:** Cutting "Access Windows" in the tower wall is **FORBIDDEN** (preserves structural integrity).
 - **Weights:** Stackable, laser-cut steel plates.
 - **Cable Routing:** The suspension rope for the bottom counterweight passes through a clearance hole in the upper counterweight.
- **Drive System (Z-Lift):**
 - **Type:** Rack & Pinion ("Climbing Motor" configuration).
 - **Mounting:** Rack fixed to the column side/face; Motor + Gearbox mounted on the moving Carriage.
 - **Actuation:** 200W Brake Servo.
 - **Balancing:** Counterweight mass matches (Carriage + SCARA Arm) mass exactly.

4.2. SCARA Arm 2 Configuration

- **Reach:** Extended reach (compared to Arm 1) if necessary, or identical 800mm standard.
- **Joint configuration:** Identical to Arm 1 (400W Shoulder / 200W Elbow / NEMA 17 Wrist) for parts commonality.
- **End Effector: Universal Lead-Screw Gripper** (Identical to Arm 1).

- **Design:** Stepper driven parallel gripper with "Hybrid Jaws".
- **Features:** Integrated rigid hook for container transfer.

4.3. Vision & Validation System

- **Cameras:** Fixed external array or arm-mounted camera pointing at the picking zone.
- **Function:** QR Code recognition and pose estimation.

5. Operational Workflows

5.1. Workflow A: Main Manipulator (Retrieval)

1. **Alignment:** Tower aligns X-axis; Z-Axis A aligns Work Surface with Drawer; Z-Axis B aligns Arm 1.
2. **Extraction:** Arm 1 rotates wrist to engage **Hook**; pulls Drawer onto Z-Axis A Work Surface.
3. **Picking:**
 - **Target:** Box or Bottle.
 - **Action:** Arm 1 rotates wrist to align Gripper Jaws with the item's face/diameter.
 - **Grip:** Jaws close (Force Controlled).
 - **Place:** Arm deposits item into Logistics Container.
4. **Return:** Arm 1 re-engages Hook and pushes Drawer back into Cabinet.

5.2. Workflow B: Sorting Station (Induction)

1. **Inbound:** Conveyor moves Items (Boxes/Bottles) past the station.
2. **Detection:** Conveyor stops when Item reaches the pick zone.
3. **Pick:** SCARA Arm 2 aligns jaws (via Wrist Rotation) and grips the item.
4. **QR Verification:**
 - Cameras attempt to read QR code.
 - *If unreadable:* Arm 2 rotates the wrist (Joint 3) or the entire arm to expose hidden sides of the Item to the camera array.
5. **Vertical Loading:**
 - Once verified, Arm 2 rotates the Item 90° (if needed) to align with the container slot.
 - Arm 2 inserts the Item **vertically** into the **Uploading Container**.
6. **Distribution (Container Handoff):**
 - Once the Uploading Container is full, the **Main X-Z Manipulator** travels to the Sorting Station and aligns its empty Container Carriage (Z-Axis A).
 - **Transfer:** The Standalone SCARA (Arm 2) engages the Uploading Container using its **Integrated Hook**.

- Arm 2 pushes/pulls the container onto the Main Manipulator's carriage (bridging the reach gap that prevents Arm 1 from grabbing it).
- Main Manipulator secures the container and distributes Items to Cabinets.

6. Design Options & Engineering Best Practices

6.1. Structural Base (X-Axis Railway)

- **Material:** Rectangular Steel Tube (200x200x6 mm or 300x200x6 mm).
 - *Rationale:* Closed box section provides superior torsional rigidity compared to open C-profiles, critical for minimizing tower twist during acceleration.
- **Base Width Strategy:**
 - **200mm Width:** Validated via calculation.
 - *Deflection:* Tip deflection calculated at **0.14 mm** under a 100N inertial load (20kg @ 5m/s² lateral acceleration). This is within acceptable limits for the application.
 - **Anchoring:** Critical. The narrow base requires the tube to be bolted to the floor at 500-1000mm intervals to manage overturning moments (where $M_{\text{overturning}} > M_{\text{stabilizing}}$).
- **Railway Assembly Strategy (Modular Build):**
 - **Objective:** Assemble a 6-8m continuous railway from 2m modular sections.
 - **Joining Method:** Butt-jointed tubes connected by heavy **Bolted Side Plates**.
 - **Alignment Protocol ("Splint & Clamp"):**
 1. Butt two 2m tube sections together. Loosely install side plates.
 2. Clamp a precision straight edge (or spare rail) across the **Top Surface** of the joint to force the tubes flush/coplanar.
 3. Tighten side plate bolts while clamped to lock the alignment.
 4. **Rail Installation:** Mount the linear rails such that the rail joint is **staggered** (offset from the tube joint (e.g., by 200mm). The rail itself acts as a final bridge to smooth the transition for the bearing blocks.

6.2. Tower Construction

Option A: Bent Sheet Metal (Recommended)

- **Material:** 4mm Carbon Steel Sheet (S235/S355). 3mm is not recommended due to flexibility.
- **Design:** CNC Bent "C" profile with a **Bolted Back Panel**.
- **Stiffness Strategy:**
 - **Stitch Bolting:** The back panel must be bolted every **50-75mm** to restore "Closed Box" torsional rigidity.

- **Internal Ribs:** 2-3 horizontal bulkhead plates welded or bolted internally to prevent cross-section distortion.
- **Base Interface:** The bottom of the tower sheet is **folded out** to form the mounting flange. This provides a clean, continuous load path and avoids complex welding/warping at the base.
 - **Leveling Strategy:** Due to CNC bending tolerance ($\pm 0.5^\circ$), the flange must be mounted with **4x Jack Bolts (Grub Screws)** under the corners to "tram" the tower vertical before tightening the main anchor bolts.
- **Internal Mechanics:** Counterweight rails/guides can be bolted directly to the inside face of the open tower before the back panel is installed. No cartridge required.

Option B: Solid Welded Tube (Alternative)

- **Material:** Solid Steel Square Tube (200x200 mm, 4-5mm wall).
- **Base Interface:** Heavy steel base plate (15-20mm) welded to the bottom.
- **Internal Mechanics:** Requires the **Drop-In Cartridge** system described in Section 4.1.
- **Prohibited:** Laser-cut truss structures are forbidden.
- **Counterweight Rail Options:**
 - **Dual Monorail Strategy (X-Profile):** Two independent counterweights running on the same X-profile spine.
 - **Guidance:** Single vertical linear rail (e.g., MGN15) mounted to the X-profile face.
 - **Constraint: Two Blocks per Weight** required (spaced 100-150mm vertically).
 - **Kinematic Suspension Strategy:**
 - **Concept:** Create a "3-Point Kinematic Mount" (2 Rail Blocks + 1 Floating Rope Point).
 - **Requirement:** The linear rail blocks must connect to the counterweight carriage **indirectly** via a **Rubber Isolator Block** or **Elastic Element** (not bolted directly).
 - **Benefit:** This elasticity absorbs minor misalignments and manufacturing tolerances, preventing the rail blocks from binding due to over-constraint while maintaining smooth linear guidance.
 - **Hybrid "Side-by-Side" Strategy (Bent Tower Only):**
 - **Layout:** Two counterweights running side-by-side (Left/Right split).
 - **Primary Guide:** One Linear Rail (MGN15/9) mounted on the **Front Face** (inside).
 - **Secondary Guide (Anti-Twist):** One Linear Shaft (10-12mm) mounted on the **Side Wall**.
 - **Benefit:** The dual-guide constraint (Rail+Shaft) allows the rope to be attached near the **tower centerline** (closer to the Carriage) rather than the weight's CoM, simplifying rope management.

6.3. Linear Rail & Rack Mounting

- **Rail Selection:** Hiwin HG25 (or equivalent).
 - **Accuracy Class:** **C** (Normal) or **H** (High). Precision (P) is unnecessary for a welded frame.
 - **Preload:** **Z0** (Light) or **ZA** (Medium). *Avoid ZB (Heavy) as it requires perfect surface flatness.*
- **Mounting Strategy ("Master & Slave"):**
 - **Step 1 (Master):** Align and bolt the first rail using a reference (laser/wire).
 - **Step 2 (Slave):** Mount the second rail loosely. Bolt the carriage to the blocks on both rails. Slide the carriage along the track; it will pull the Slave rail into parallel alignment with the Master. Tighten Slave bolts progressively.
- **Rack Mounting:**
 - **Configuration:** Rack is bolted to the side face of the tube (or top face with side teeth). Pinion axis is vertical (Z-axis).
 - **Spacer:** A **2mm Laser-Cut Shim Plate** is installed between the rack and the tube surface. This lifts the gearing off the rough structural steel to prevent pinion rubbing and provides a clean seating surface.

6.4. Maintenance Strategy

- **Lubrication:**
 - **Grease Type:** Lithium Soap Base Grease No. 2 (Standard industrial).
 - **Interval:** Every **3-6 months** or **100km** of travel.
 - **Method:** Manual greasing via standard grease gun (M6/PT1/8 nipple on blocks). Automated systems are not required for this duty cycle.
- **Alignment & Life:** Proper rail parallelism ($e_1 \approx 0.025mm$) is critical. Exceeding this creates "Internal Preload," which can reduce bearing fatigue life by >50%.

6.5. Mechanical Transmission (Ropes & Pulleys)

- **Rope Selection:**
 - **Material:** **AISI 304 or 316 Stainless Steel.**
 - **Construction:** **7x19** (7 strands of 19 wires). This high-flexibility construction is critical for fatigue resistance when running over pulleys. *Avoid 1x19 or 7x7 stiff ropes.*
 - **Size:** **Ø3 mm.** (Safety Factor > 5:1 for 30kg load).
- **Pulley Design & Sizing:**
 - **Critical Ratio (D/d):** The ratio of Pulley Diameter (D) to Rope Diameter (d) must be at least **25:1** to prevent premature wire fatigue.
 - **Calculation:** $D_{min} = 25 \times 3 \text{ mm} = \mathbf{75 \text{ mm.}}$

- **Specification:** Use **Ø75 mm Metal Pulleys** (Root Diameter) with ball bearings.
- **Space Constraint Note:** If the cartridge design forces a smaller pulley (e.g., Ø50mm), the rope diameter **MUST** be reduced to **Ø2 mm** to maintain the 25:1 ratio ($25 \times 2 = 50$). Do not run Ø3mm rope on a Ø50mm pulley.
- **Termination:**
 - **Method:** Thimble Loop secured with Swage Sleeves.
 - **Connection:** Connects to the **Floating Suspension** point (Rubber Isolator) on the counterweight carriage via a threaded rod for length adjustment.

6.6. Electrical & Static Safety

- **Triboelectric Risk:** High-speed plastic rollers, belts, and cable chains generate significant static charge. Without a discharge path, this will arc to the nearest ground (electronics), causing faults.
- **Grounding Strategy ("Star Ground"):**
 1. **Stationary Base:** Tower and Rail Base must be bolted to Earth Ground.
 2. **Moving Carriage:** Must be bonded to the Tower via a **Flat Braided Ground Strap** ([Link: McMaster](#)) routed inside the cable chain.
 3. **Counterweights:** If using non-conductive rollers/bushings, the counterweight is electrically isolated.
 - **Requirement:** Install a **Conductive Brush** (Carbon or Copper Beryllium) on the counterweight carriage that continuously wipes against the grounded steel tower wall or rail.
 4. **End Effector:** Gripper jaws/cups must be ESD safe (conductive rubber) to prevent static buildup on picked items.
- **Belts:**
 - **Primary:** Use **Antistatic Timing Belts** (e.g., [BrecoFlex](#) or [Gates PowerGrip](#)).
 - **Secondary Discharge (Required):** Install **Passive Discharge Brushes** (Carbon Fiber or Tinsel) positioned 1-2mm from the belt surface near the pulleys ([Link: Fraser Anti-Static](#) or [McMaster Static Brushes](#)).
 - *Reason:* Even antistatic belts lose surface conductivity over time due to wear; brushes provide fail-safe discharge.
- **Rail & Shaft Grounding:**
 - **Carriage Wiper:** Use **Carbon Brushes** ([Link: Schunk](#) or [Mersen](#)) or **Beryllium Copper Fingerstock** ([Link: DigiKey](#) or [Tech-Etch](#)).
 - **Shaft Grounding:** Use **AEGIS Rings** ([Link: AEGIS](#)) or **Conductive Grease** ([Link: Kluber](#)).

6.7. SCARA Transmission & Drive

- **Belt Profile Selection:**
 - **Primary Recommendation (Production): Gates PowerGrip GT3 / GT2 (5M Pitch).**
 - *Benefit:* Zero backlash, high precision.
 - **MVP / Rapid Prototype Selection: HTD (High Torque Drive) 3M or 5M.**
 - *Rationale:* Allows use of readily available **Aluminum Timing Bar Stock** for custom pulley fabrication.
 - *Backlash Analysis:* HTD profile has inherent play (~0.1mm linear). Cumulative error at arm tip calculated at **~1.5mm**.
 - *Verdict:* **Acceptable** for the 3-5mm picking precision target of the MVP.
- **Pulley Standardization Strategy:**
 - **Objective:** Use identical pulley stock for both Shoulder and Elbow joints to simplify inventory.
 - **Small Pulley (Drive): 20 Teeth** (5M Pitch).
 - Pitch Diameter: 31.83 mm.
 - **Large Pulley (Driven): 72 Teeth** (5M Pitch).
 - Pitch Diameter: 114.59 mm.
 - **Resulting Belt Ratio: 3.6:1.**
- **Gearbox & Reduction Selection Matrix:**
 - The total reduction is derived from $\text{Gearbox Ratio} \times \text{Belt Ratio (3.6:1)}$.
 - **Belt Stage is fixed** at 3.6:1 (20T Drive / 72T Driven) to standardize parts.

Gearbox Ratio	Total Reduction	Application Use Case	Max Torque Potential (at Arm)	Speed @ 3000 RPM Motor
5:1	18:1	High Speed (Elbow J2). Use for ultra-fast cycle times (<0.5s). Lower stiffness.	Medium	~166 RPM (Very Fast)
7:1	25.2:1	Balanced (Elbow J2). Good compromise for general picking.	High	~119 RPM (Fast)
10:1	36:1	High Torque / Stiffness (Shoulder J1). Recommended for heavy loads/long reach.	Very High	~83 RPM (Standard)
15:1	54:1	Max Precision (Shoulder J1). Use if 10:1 stiffness is insufficient.	Extreme	~55 RPM (Slow/Precise)

- **Final Selection Recommendation:**

- **Standardization:** Both Joints (J1 and J2) use **10:1 Gearboxes** and **20T/72T Pulleys**.
- **Belt Width: 25mm** for both Shoulder and Elbow.
 - *Rationale:* While J2 only needs 15mm, using 25mm everywhere eliminates the need to stock two different belt widths and pulley blanks. The weight penalty (~200g) is negligible compared to the logistical benefit.
- **Wrist Joint Transmission (J3):**
 - **Motor:** NEMA 17 Stepper (Low torque, low inertia).
 - **Belt Profile:** **GT2/GT3 3M Pitch** (3mm) or **HTD 3M** (MVP). *3M is recommended for compact pulleys.*
 - **Belt Width: 6mm or 9mm.**
 - **Pulley Sizing:**
 - **Drive:** 20 Teeth (3M).
 - **Driven:** 60 Teeth (3M).
 - **Ratio: 3:1** (Target speed/torque balance for "flipping" action).
 - **Note:** Wrist transmission is intentionally different from J1/J2 to optimize for low mass and compact packaging at the end effector.

6.8. Materials Guide: Belts & Pulleys

- **Belt Body Material:**
 - **Polyurethane (PU):** White/Clear.
 - *Pros:* Very stiff (low stretch), clean, abrasion-resistant.
 - *Cons:* Can generate static. **Must specify Antistatic/Conductive PU.**
 - *Verdict:* **Recommended** for SCARA rigidity.
 - **Neoprene (Rubber):** Black.
 - *Pros:* Quieter, naturally more conductive (antistatic), handles shock loads better.
 - *Cons:* Slightly more "bouncy" (compliant) than PU.
 - *Verdict:* Acceptable, but PU is preferred for positioning stiffness.
- **Tensile Cord Material:**
 - **Fiberglass:** Standard. *Too much stretch for SCARA. Avoid.*
 - **Steel:** High stiffness. *Standard for automation. Recommended.*
 - **Kevlar/Aramid:** Light, no rust. *Good, but steel is stiffer.*
 - **Carbon Fiber:** Ultra-stiff. *Expensive, but best for high-speed settling.*
- **Pulley Width Sizing Rule:**
 - The pulley face width must be wider than the belt to allow for tracking.
 - **Rule:** Pulley Face = Belt Width + 15% to 20% .

- **Example (J1 & J2):** 25mm Belt → **30mm - 32mm** Pulley Face.

6.9. Rotary Joint Bearings

- **Production Recommendation (High Performance):**
 - **Type:** Cross Roller Bearing (RU Series).
 - **Size:** RU85 (ID 55mm / OD 120mm) for both joints.
 - **Rationale:** Best stiffness and torque capacity.
- **MVP / Cost-Optimized Option (Recommended for 3D Print/Laser Cut):**
 - **Architecture:** "Dual Ball Bearing Sandwich".
 - **Concept:** Instead of one expensive CRB, use two standard deep groove ball bearings separated vertically to handle the moment load.
 - **Bearings:** **6808-2RS** (Thin Section, 40mm ID x 52mm OD) or **6008-2RS** (Standard, 40mm ID x 68mm OD).
 - *Cost:* ~\$10-15 per joint vs \$80+ for RU85.
 - *Availability:* Universal standard part.
 - **Assembly:**
 - **Structure:** Top Plate and Bottom Plate (Laser Cut Steel) act as housing.
 - **Shaft:** **40mm OD Hollow Aluminum/Steel Tube** passes through the joint (ID ~34-35mm available for cables).
 - **Preload:** A threaded nut or clamp on the central tube pulls the inner races together against a spacer, removing play.
 - **Precision:** Adequate for 3-5mm pick accuracy with a 400mm+400mm arm.

6.10. End Effector: Universal Lead-Screw Gripper

- **Design Concept:** A single, adaptive tool that handles both thin boxes and wide bottles using a wide stroke range and force control, eliminating the need for complex tool changers or flipping heads.
- **Actuator:**
 - **Motor:** NEMA 17 Stepper.
 - **Drive:** **Right-Hand / Left-Hand T8 Lead Screw** (Standard 3D printer component).
 - **Mechanism:** Dual-carriage self-centering action. One screw moves both jaws symmetrically.
 - **Stroke Range:** **0mm to 100mm+** (Enables gripping 8mm boxes and 80mm bottles).
- **"Hybrid Jaw" Geometry (Custom 3D Printed Fingers):**
 - **Zone A (Tips):** Flat Rubber Pads.
 - *Function:* Grips the **Side Faces** of thin boxes (e.g., medicine packs). The thin jaw tips slide into the drawer compartment gaps.

- **Zone B (Middle): V-Groove.**

- *Function:* Grips cylindrical items (bottles). The V-shape centers the bottle and provides a secure 4-point lock.